

About 2511.20527

jhy

December 7, 2025

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1 Conventional Abelian Higgs Model vs. Degenerate Abelian Higgs Model

$$\text{Conventional AHM/Bulk string: } V_4(\phi) = \frac{\lambda}{2} (|\phi|^2 - v^2)^2 \quad (1.1)$$

$$\text{Degenerate AHM/Domain wall string: } V_6(\phi) = \frac{\lambda}{2} |\phi|^2 (|\phi|^2 - v^2)^2$$

In degenerate AHM, there are two vacua: $\langle \phi \rangle = v$ and $\langle \phi \rangle = 0$, which can be connected by a domain wall with boundary condition $\phi(-\infty) = 0, \phi(\infty) = v$, The superpotient is $W(\phi)$, defined by $V_6(\phi) = |W'(\phi)|^2$. $W'(\phi) = \sqrt{\frac{\lambda}{2}} \phi(v^2 - \phi^2)$ The domain wall tension in

$$\begin{aligned} \sigma_{DW} &= \int dz (\phi'(z)^2 + W'(\phi)^2) \\ &= \int dz \left((\phi'(z) - W'(\phi))^2 + 2\phi'(z)W'(\phi) \right) \\ &\geq 2 \int dW' \\ &= 2(W(v) - W(0)) = \frac{\sqrt{\lambda}}{2\sqrt{2}} v^4 \end{aligned} \quad (1.2)$$

The condition for equality is

$$\phi'(z) = W'(\phi) \Rightarrow \phi(z) = \frac{v}{\sqrt{1 + e^{-\sqrt{2\lambda}v^2(z-z_0)}}} \quad (1.3)$$

2 Magnetic vortices

$$A_0 = A_r = 0, A_\theta = A_\theta(r) \quad (2.1)$$

$$T = \int d^2x \left(\frac{1}{4} (F_{\mu\nu})^2 + |D_\mu \phi|^2 + V(\phi) \right) \quad (2.2)$$

Profile functions:

$$\begin{aligned} A_\theta &= n(1 - a(r)), \quad \phi = v\varphi(r)e^{in\theta} \\ \varphi(r=0) &= 0, \varphi(r=\infty) = 1, a(r=0) = 1, a(r=\infty) = 0 \end{aligned} \quad (2.3)$$

We introduce a dimensionless radial coordinate

$$u = evr, \quad \tilde{V}(\varphi) = \frac{1}{v^4 e^2} V(v\varphi), \quad \beta = \lambda v^2 / e^2 \quad (2.4)$$

$$\frac{T_n}{2\pi v^2} = \int_0^\infty u du \left(\frac{n^2}{2u^2} a'(u)^2 + \varphi'(u)^2 + \frac{n^2}{u^2} a^2(u) \varphi^2(u) + \tilde{V}(\varphi(u)) \right) \quad (2.5)$$

Then the vortex equations are

$$\begin{aligned} \varphi''(u) + \frac{1}{u} \varphi'(u) &= \frac{n^2}{u^2} a^2(u) \varphi(u) + \frac{1}{2} \frac{d\tilde{V}}{d\varphi} \\ a''(u) - \frac{1}{u} a'(u) &= 2a(u) \varphi^2(u) \end{aligned} \quad (2.6)$$

3 Asymptotical Analysis of the Vortex Equations

i.

$$\begin{aligned} u \rightarrow 0 &\Rightarrow \tilde{V} \rightarrow 0, a \sim 1 \\ &\Rightarrow \varphi''(u) + \frac{1}{u} \varphi'(u) \sim \frac{n^2}{u^2} \varphi(u) \quad \text{which is an Euler equation,} \\ &\Rightarrow \varphi(u) \sim c_n u^n \\ &\Rightarrow a''(u) - \frac{1}{u} a'(u) \sim 2c_n^2 u^{2n} \\ &\Rightarrow a = A + Bu^2 + \frac{c_n^2}{2n(2n+1)} u^{2n+2} \sim 1 + Bu^2 \equiv 1 - \frac{u^2}{u_n^2} \\ &\Rightarrow \frac{1}{e^2 v^2} F_{12} \equiv B(u) = -\frac{n}{u} a'(u) = \frac{2n}{u_n^2} \end{aligned} \quad (3.1)$$

ii.

$$\begin{aligned} u \rightarrow \infty &\Rightarrow a''(u) - \frac{1}{u} a'(u) = 2a(u) \quad \text{which is a modified Bessel equation} \\ &\Rightarrow a(u) \sim a_\infty \sqrt{u} e^{-\sqrt{2}u} \\ &\Rightarrow \varphi''(u) + \frac{1}{u} \varphi'(u) \sim \frac{n^2}{u} a_\infty^2 e^{2\sqrt{2}u} \\ &\Rightarrow \varphi = 1 - \frac{\varphi_\infty}{\sqrt{u}} e^{-\sqrt{2\beta}u} - \frac{n^2 a_\infty^2 e^{-2\sqrt{2}u}}{2(\beta-4)u} \end{aligned} \quad (3.2)$$

if $\beta < 4$

$$\varphi(u) \sim 1 - \frac{\varphi_\infty}{\sqrt{u}} e^{-\sqrt{2\beta}u} \quad (3.3)$$

if $\beta > 4$

$$\varphi(u) \sim 1 - \frac{n^2 a_\infty^2 e^{-2\sqrt{2}u}}{2(\beta - 4)u} \quad (3.4)$$

Note that $m_H r = \sqrt{2\beta}u, m_V r = \sqrt{2}u$.

4 If $n \rightarrow \infty$

If $n \rightarrow \infty$, the magnetic flux will become infinite, so that the domain where $a = 1$ will become infinitely large, namely

$$u_n \rightarrow \infty \quad (4.1)$$

This point is obtained using numerical methods in the paper. Roughly speaking, we can consider u_n as the radius of the domain of Coulomb phase. And

$$a(u < u_n) \sim 1 - \frac{u^2}{u_n^2}, a(u > u_n) \sim 0, \varphi(u < u_n) \sim 0, \varphi(u > u_n) \sim 1 \quad (4.2)$$

5 Bulk Strings vs. Domain Wall Strings

$$\mathcal{O}(u_n) = \text{domain wall term} \quad (5.1)$$

$$\mathcal{O}(u_n^2) = \text{bulk term}$$

$$\sigma(\beta) = \frac{\sigma_{DW}}{2^{1/4}e} \quad (5.2)$$

5.1 Large- n Bulk Strings

$$\frac{T_{n,core}}{2\pi v^2} \sim \int_0^{u_n} u du \left(\frac{2n^2}{u_n^4} + \frac{\beta}{2} \right) = \frac{n^2}{u_n^2} + \frac{\beta}{4} u_n^2 \quad (5.3)$$

The magnetic flux contributes to the first term, and the Higgs potential $\tilde{V}_4$ the second. Then

$$\frac{\partial T_n}{\partial u_n} = 0 \Rightarrow u_n(\beta) = \frac{\sqrt{2n}}{\beta^{1/4}} \quad (5.4)$$

In the above discussion we have neglected the boundary region around $u \simeq u_n$ that separates the core region of the string from the Higgs vacuum in the exterior region.

$$T_{n,boundary} \sim (2\pi u_n) \sigma(\beta) \quad (5.5)$$

Then

$$u_n^{imp} = \frac{\sqrt{2n}}{\beta^{1/4}} - \frac{\sigma}{2\sqrt{2}\beta} \quad (5.6)$$

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$$\begin{aligned} \varphi''(u) + \frac{1}{u} \varphi'(u) &= \frac{n^2}{u^2} a^2(u) \varphi(u) + \beta \varphi(u) (\varphi^2(u) - 1) \\ a''(u) - \frac{1}{u} a'(u) &= 2a(u) \varphi^2(u) \end{aligned} \quad (5.7)$$

|||||||

5.2 Large-n Domain Wall Strings

$$\frac{T_{n,core}}{2\pi v^2} = \frac{n^2}{u_n^2} \quad (5.8)$$

$$T_{n,boundary} \sim (2\pi u_n)\sigma(\beta) \quad (5.9)$$

$$\frac{\partial T_n}{\partial u_n} = 0 \Rightarrow u_n = \frac{2^{1/3}v^{2/3}n^{2/3}}{\sigma^{1/3}} \quad (5.10)$$

Now we know that if $u_n \propto n^{1/2}$, it is a bulk string; and if $u_n \propto n^{2/3}$, it is a domain wall string. In the large-n limit, the leading order of T_n contributes the most, so bulk string: $T_n \propto u_n^2 \sim n$, domain wall string: $T_n \propto u_n \sim n^{2/3}$.

6 Stability

$$\begin{aligned} \frac{d^2 T_n}{dn^2} < 0 & : stable \\ \frac{d^2 T_n}{dn^2} > 0 & : unstable \end{aligned} \quad (6.1)$$

Bulk string:

$$\frac{1}{2\pi v^2} \frac{d^2 T_n}{dn^2} = \frac{\sigma}{n^{3/2}\beta^{1/4}} \left(\frac{1}{4\beta} - \frac{1}{2\sqrt{2}v^2} - \frac{1}{16} \right) \quad (6.2)$$

Domain wall string:

$$\frac{1}{2\pi v^2} \frac{d^2 T_n}{dn^2} = -\frac{2^{1/3}\sigma^{2/3}n^{-4/3}}{3v^{4/3}} < 0 \quad (6.3)$$

7 Interaction

Yukawa potential?